

OpenClaw AI Rock Paper Scissors Game Tutorial

Before starting, please complete [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], and [05-Three Dialogue Configuration Tests] in the `Preparation Before Use` directory. If you are only using TUI/WhatsApp, you can skip `04`. This article assumes the basic environment is ready and only covers the AI rock paper scissors project itself.

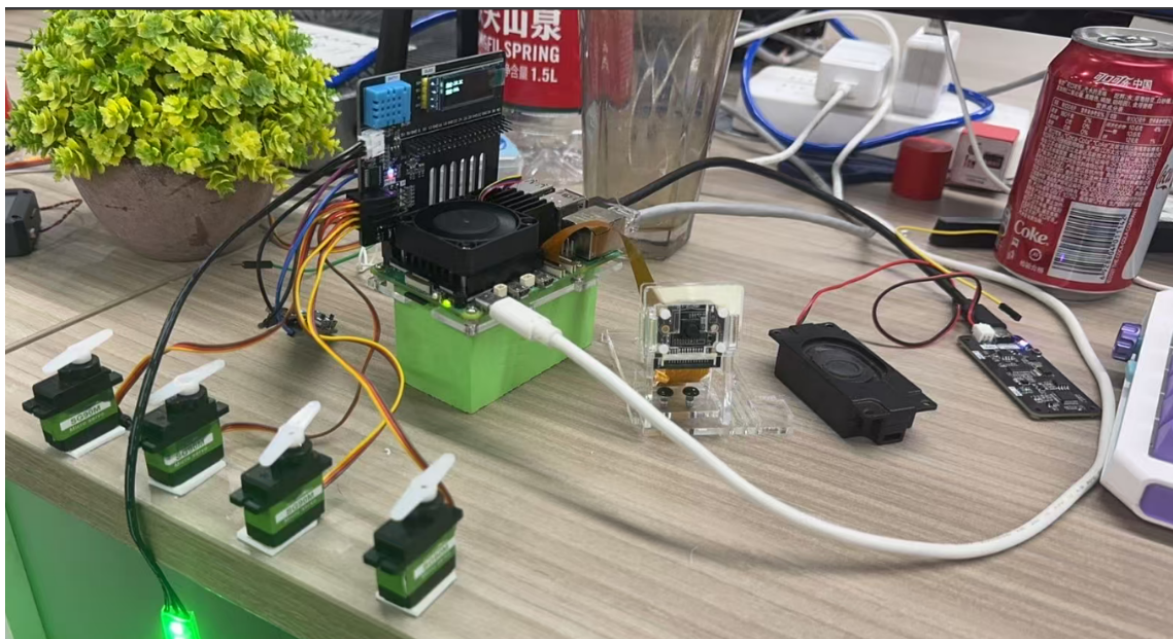
1. Tutorial Objectives

This article demonstrates OpenClaw's interactive game capability through "gesture recognition + AI referee + screen light feedback". After completing this tutorial, readers will be able to:

- Understand how to use cameras to recognize player gestures
- Communicate with OpenClaw via WhatsApp, TUI, and voice commands
- Complete four project scenarios: gesture recognition, AI move, result judgment, score feedback

2. Hardware Preparation

- OpenClaw main control board
- CSI camera module
- OLED display screen
- RGB light strip / RGB LED module
- [Optional] AI voice module + speaker
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial:

- Camera capture command: `/home/pi/hardware_ha1_demo/camera`

- Visual analysis script: `/home/pi/.openclaw/workspace/skills/pi-vision/scripts/csi_describe`
- Voice entry: `/home/pi/voice_to_openclaw/src/main.py`

The key capabilities of this project are mainly three parts:

- Camera is responsible for capturing player gestures
- `csi_describe` gesture is responsible for recognizing gestures
- OpenClaw is responsible for deciding AI moves and displaying results on OLED/RGB

4. Testing Project Basic Capabilities

Before starting this project, please complete an `openclaw tui` basic conversation self-check first; if you haven't done this yet, see [05-Three Dialogue Configuration Tests]. After passing, use TUI to separately verify the basic capabilities in the project.

Input in terminal:

```
openclaw tui
```

```

pi@raspberrypi: ~
File Edit Tabs Help
pi@raspberrypi:~ $ openclaw tui
[11] OpenClaw 2026.3.12 (6472949)
  Your terminal just grew claws-type something and let the bot pinch the busywork.

openclaw tui - ws://127.0.0.1:18789 - agent main - session main
connecting | idle
agent main | session main | unknown | tokens ?

18:28:21 [plugins] feishu_chat: Registered feishu_chat, feishu_chat_members
18:28:21 [plugins] feishu_im: Registered feishu_im_user_message, feishu_im_user_fetch_resource, feishu_im_u
ser_get_messages, feishu_im_user_get_thread_messages, feishu_im_user_search_messages
18:28:21 [plugins] feishu_search: Registered feishu_search_doc_wiki
18:28:21 [plugins] feishu_drive: Registered feishu_drive_file, feishu_doc_comments, feishu_doc_media
18:28:21 [plugins] feishu_wiki: Registered feishu_wiki_space, feishu_wiki_space_node
18:28:21 [plugins] feishu_sheets: Registered feishu_sheet
18:28:21 [plugins] feishu_im_bot_image: Registered feishu_im_bot_image
18:28:21 [plugins] Registered all OAPI tools (calendar, task, bitable, search, drive, wiki, sheets, im)
18:28:21 [plugins] feishu_doc: Registered feishu_fetch_doc, feishu_create_doc, feishu_update_doc
18:28:21 [plugins] feishu_oauth: Registered feishu_oauth tool

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | baillian/qwen3.5-plus | tokens 0/1.0m (0%)

```

Then you can first do a few basic tests:

- Take a photo of my gesture for me
- Recognize if this is a scissors hand
- Display "Ready" on OLED
- Make RGB light up blue

5. Three Ways to Communicate with OpenClaw

Method	Suitable Scenario	Advantages	Suggested Use
WhatsApp	Remote demonstration, event interaction	Easy to share for multi-person experience	Demonstrate "visual interactive game"
TUI	Local debugging, rule verification	Easiest to see recognition and judgment process	First to get the game logic working
Voice Code	Game interaction, classroom demonstration	More like a real AI game partner	As the complete demonstration solution

If you have the optional AI voice module + speaker, the TUI and WhatsApp conversations need to broadcast OpenClaw's replies. You can run the following program in the terminal. This is not needed for AI voice module conversations.

- Modify the TTS enable parameter

```
/home/pi/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast replies. After saving, run the program below in the terminal.
```

- Terminal input

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) pi@raspberrypi:~/voice_to_openclaw $ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/pi/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [05-Three Dialogue Configuration Tests]. After creating, you can remotely trigger the rock paper scissors game in WhatsApp, for example:

- Let's play rock paper scissors. Take a photo to recognize my gesture. You can randomly make a gesture. Then judge whether you win or I win. OLED displays who won this round

20:27



< 2

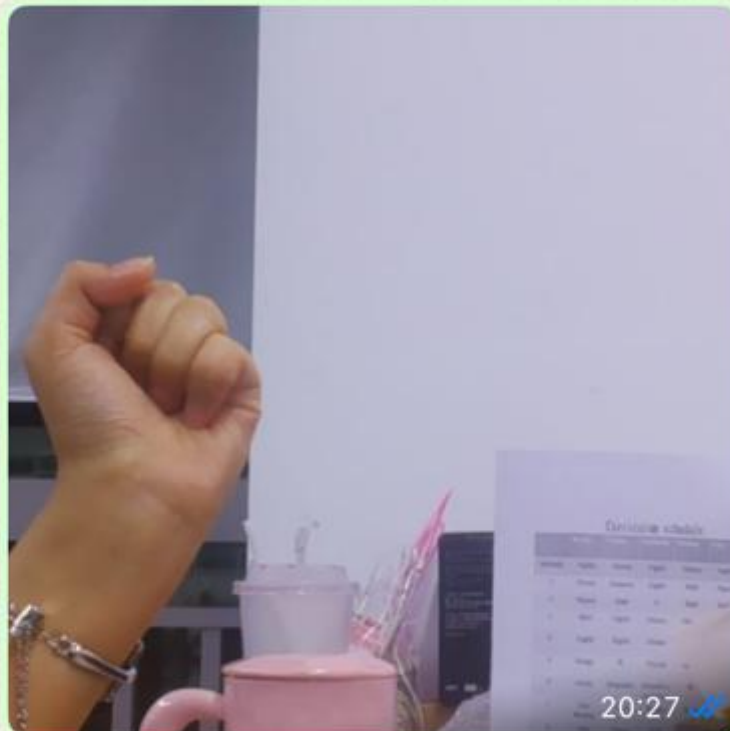


+86 153 3885 7526 (你)

给自己发消息

Please play rock paper scissors.
Take a photo to recognize my
gesture. You can randomly make a
gesture. Then judge whether you win
or I win OLED displays who this round.

20:27 ✓✓



I can see the user's hand gesture in
the image - it's a **fist**, which
represents **ROCK** in rock paper
scissors.

Now I need to:

1. Make a random gesture choice
(rock, paper, or scissors)
2. Determine the winner
3. Display the result on OLED



Here you can add WhatsApp conversation screenshots yourself.

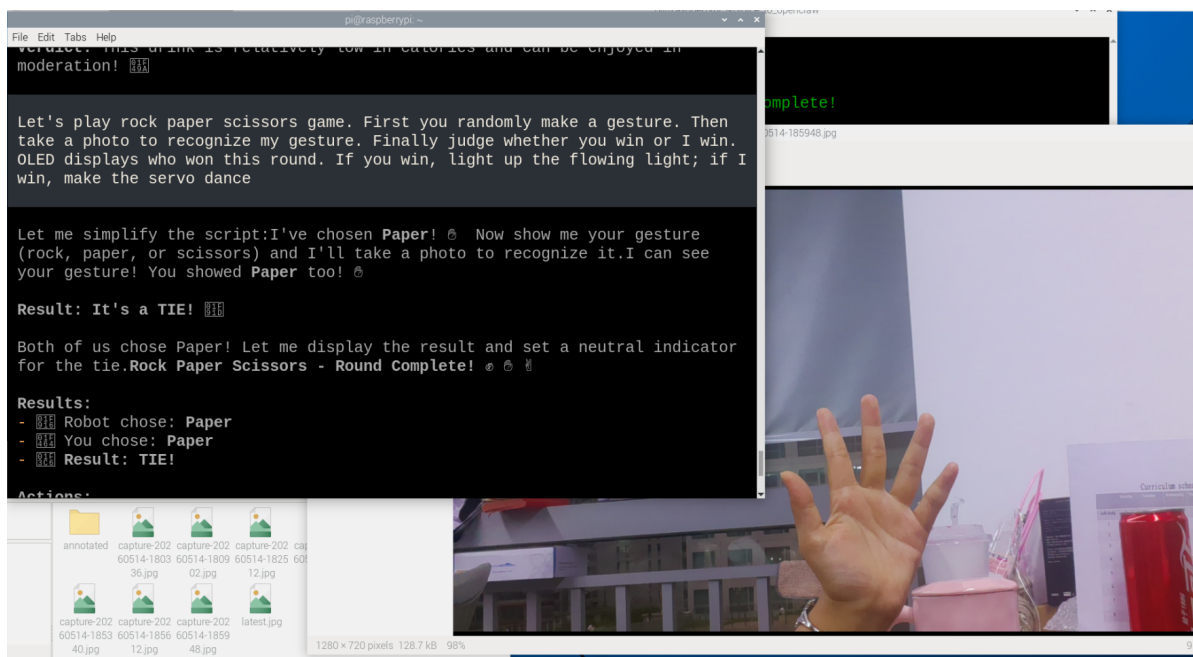
5.2 TUI Solution

If you have already completed the TUI self-check from chapter 0, directly enter terminal text dialogue:

```
openc1aw tui
```

Then input project-related commands, for example:

- Let's play rock paper scissors game. First you randomly make a gesture. Then take a photo to recognize my gesture. Finally judge whether you win or I win. OLED displays who won this round. If you win, light up the flowing light; if I win, make the servo dance

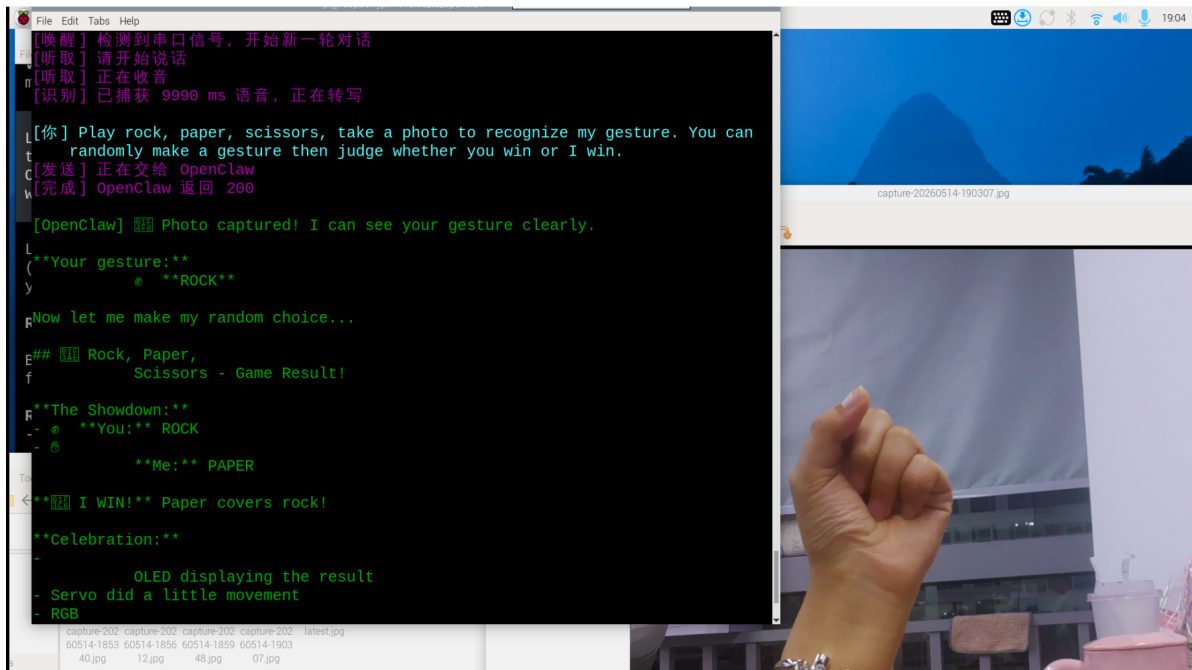


5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up flow, please refer directly to [04-AI Voice Interaction Configuration]. After waking up, you can directly speak control content, for example:

- Let's play rock paper scissors. Take a photo to recognize my gesture. You can randomly make a gesture. Then judge whether you win or I win. OLED displays who won this round



6. Project Implementation

Note: All demonstrations below use the `openclaw tui` method for conversation. You can also choose WhatsApp or voice to complete.

6.1 Scenario 1: Gesture Recognition

Experiment Objective

Recognize the player's current gesture, such as rock, scissors, or paper.

Effect Description

OpenClaw will first take a photo, then recognize the gesture type based on the image. This step is the input foundation of the entire game.

Example Commands

- Recognize my current gesture
- I've already made my move, help me check
- See if this is rock, scissors, or paper

6.2 Scenario 2: AI Move and Judgment

Experiment Objective

Have OpenClaw, after recognizing the player's gesture, give the AI's move result and determine win/loss.

Effect Description

This phase most reflects the characteristics of "visual input + dialogue reasoning". OpenClaw not only recognizes the player but also continues to participate in the game, giving its own move and result explanation.

Example Commands

- Let's play rock paper scissors. Take a photo to recognize my gesture. You can randomly make a gesture. Then judge whether you win or I win. OLED displays who won this round

6.3 Scenario 3: OLED Scoreboard

Experiment Objective

Have OLED display the current round result or cumulative score.

Effect Description

The screen can display `win`, `Lose`, `Draw`, or also display simple scores like `You 1 : 0 AI`, making the project more like a complete mini-game.

Example Commands

- Display this round's result on OLED
- Write the score to the screen
- Display "Round 1 You Win"

6.4 Scenario 4: RGB Result Feedback

Experiment Objective

Quickly express win/loss results with RGB colors.

Effect Description

For example, light up green for win, red for loss, blue for draw. This way, even without looking at text, you can immediately know the result of this round.

Example Commands

- Light up green if I win
- Light up red if I lose
- If you win, light up the flowing light; if I win, make the servo dance